Natural Remanent Magnetizations of Some Intrusive Rocks from the Sør Rondane Mountains, Queen Maud Land, Antarctica

J. D. A. ZIJDERVELD

Paleomagnetisch Laboratorium der Rijksuniversiteit Utrecht
Fort Hoofddijk, Utrecht, Netherlands

Eighteen samples from three different intrusive rocks from the Sør Rondane Mountains (East Antarctica) contained a NRM with consistent directions around 341°, +64°. This result is similar to the NRM direction of the gneisses from the Ongul Island and Lützow-Holm Bay, which belong to the same complex. It is argued that the NRM dates from the latest thermal event in this region, which is indicated by the Rb-Sr age of the biotites of the rocks as being 480 m.y. for the Sør Rondane intrusives. If it is assumed that the NRM reflects the local geomagnetic field direction in the Lower Ordovician, some tentative conclusions for the position of Antarctica in Gondwanaland can be advanced.

INTRODUCTION

Since little is known of the paleomagnetism of Antarctica, the author joined the summer party to the Sør Rondane Mountains during the second Belgian-Netherlands Antarctic Expedition (1965), as organized by the Belgian-Netherlands Antarctic Committee. The Sør Rondane Mountains are situated in East Antarctica, at the border of the polar plateau, and are approximately 200 km south of the 'Base Roi Baudouin,' constructed by the 1958 Expédition Antarctique Belge (leader G. de Gerlache). During January 1965 only the western part of this 250 km long and 90 km wide mountain range could be visited. This area lies at 72°S, 24°E.

The Sør Rondane Mountains are a gneissic complex composed of an important series of gneisses and migmatites, containing marble beds, graphite-schists, and quartzites. It is a Precambrian sedimentary sequence, isoclinally folded and strongly metamorphosed in the upper catazone, with varying anatexitic phenomena and with a slight retrograde metamorphism in the mesozone [*Michot and Klerkx*, 1966]. Younger and nonmetamorphic sediments are not found [*van Autenboer et al.*, 1964]. These metamorphic rocks appeared to be very unsuitable for a preliminary paleomagnetic research.

The metamorphic complex is invaded, however, by younger igneous massifs and dykes, many of which have been dated by radiometric methods [*Picciotto, Deutsch, and Pasteels*, 1964, 1966], and these dated intrusive rocks were used to supply the paleomagnetic data. This post-orogenic and post-metamorphic igneous complex consists mainly of granites, syenites, and gabbroic rocks.

RADIOMETRIC AGES

Picciotto et al. [1964, 1966] have determined the ages of nineteen samples, representing the principal types of rocks from the region, using various methods of isotopic geochronometry. The results are summarized in Table 1. Their collection included two samples of rocks that were used in the present paleomagnetic study (granite from Romnæsfjellet and syenite from Lunckerygge).

From five of the samples additional K-Ar whole-rock measurements were made by Ravich and Krylov in Leningrad; the ages obtained were from 350 to 500 m.y.

Picciotto concluded that the emplacement of the intrusive complex, based on the age of its zircons, took place approximately 530 m.y. ago and that the metamorphism of the gneiss and the migmatization would be approximately 550 m.y. old and thus nearly contemporaneous with the magmatic intrusions. The 550-m.y. age is based on the Rb-Sr whole-rock measurements of two fine-grained anatexitic granites, considered as products of the principal migmatic phase and synchronous with the tectonic deformation.

The low K-Ar ages were determined on granitic whole rocks and are thus undoubtedly a

TABLE 1. Isotopic Ages from the Sør Rondane Mountains [*Picciotto, Deutsch, and Pasteels*, 1964]

Method	Veins, m.y.	Intrusive Complex, m.y.	Gneiss and Migmatites, m.y.
Sr/Rb on biotite	474 ± 10 (3 samples)	472 ± 7 (5 samples)	486 ± 5 (9 samples)
Sr/Rb on muscovite			510 ± 15 (1 sample)
Sr/Rb on feldspar		480 ± 160 (1 sample)	470 ± 60 (2 samples)
Sr/Rb on whole rock			550 ± 50 (2 samples)
Pb/U on zircon	505 ± 25 (1 sample)	510 to 540 or slightly older (2 samples)	Between 520 and 700 (1 sample)

consequence of argon loss. Indeed, the only diorite sample with low potassium feldspar and high biotite content revealed a K-Ar whole-rock age of 500 m.y.

The Rb-Sr ages on the biotites of all rocks, the gneisses as well as the younger intrusives, lay between 460 and 510 m.y., the mean being 480 m.y. Michot attributed this biotite Rb-Sr age plateau to the retrograde metamorphism observed in the entire metamorphic area of this part of Antarctica between 2° and 60°E. He, moreover, correlated this retromorphism with the final phase of the consolidation of the eruptive masses [*Michot and Klerkx*, 1966]. In the Sør Rondane region this slight retrograde metamorphism took place in the mesozone.

Another interpretation of the observations might connect this second metamorphic event with the intrusion of the igneous masses themselves, both events being approximately 480 m.y. old. Since the igneous rocks contain xenoliths of the gneiss complex, this could also be the source of its older zircons.

Nevertheless, for the paleomagnetic study only the final thermal event in the rocks recorded by their biotite Rb-Sr age is important.

Sampling Localities

The post-metamorphic igneous rocks are found as huge massifs and dykes of all sizes intruding the metamorphic series in the mountains, but they are also found as isolated nunataks to the north of the mountain range.

Because of the restricted conditions of the summer party, no extensive paleomagnetic collecting was possible. Each of the sampled intrusives was collected at only one locality. The sampling sites were chosen by van Autenboer, the local geologist and leader of the summer party. A detailed geologic description of the sampling region is given by *van Autenboer and Loy* [1966]. A total of eight intrusives were sampled, their locations are indicated on Figure 1 by numbers that refer to

North of the mountain range:

1. On Seal nunatak. Coarse-grained granite (7 samples and 1 sample of light colored gneiss inclusion). This granite is identical to that of the nearby large nunatak Romnæsfjellet.
2. On the eastern slope of Vesthaugen nunatak.
   2.1 Massive monzonite (4 samples), intercalating gneiss.
   2.2 Kersantite dyke (4 samples), cutting the monzonite.
3. On nunatak 1550. Thin basalt sill (7 samples), intervening gneiss.

In the mountains:

4. Syenite from Lunckerygge (7 samples), on the eastern slope of the Jenningsbreen valley.
5. Granitic dyke in syenite from Lunckeryggen (5 samples), to the southeast of the Luncke pass (western slope of the Gjelbreen valley).
6. Granite from Lunkerygge on the eastern slope of the Jenningsbreen valley.
   6.1 Granite (5 samples).
   6.2 Older aplite dyke (1 sample) in the granite.
   6.3 Younger aplite dyke (3 samples) in the granite.
7. Granite from Vengen (9 samples) in the west wall of the Vengen mountain, between hill 2500 and hill 2550.
8. Hornblende-epidote rock (probably metamorphosed diabase) in the east wall of Widerøefjellet (6 samples).

The local geomagnetic declination was 35.5°W

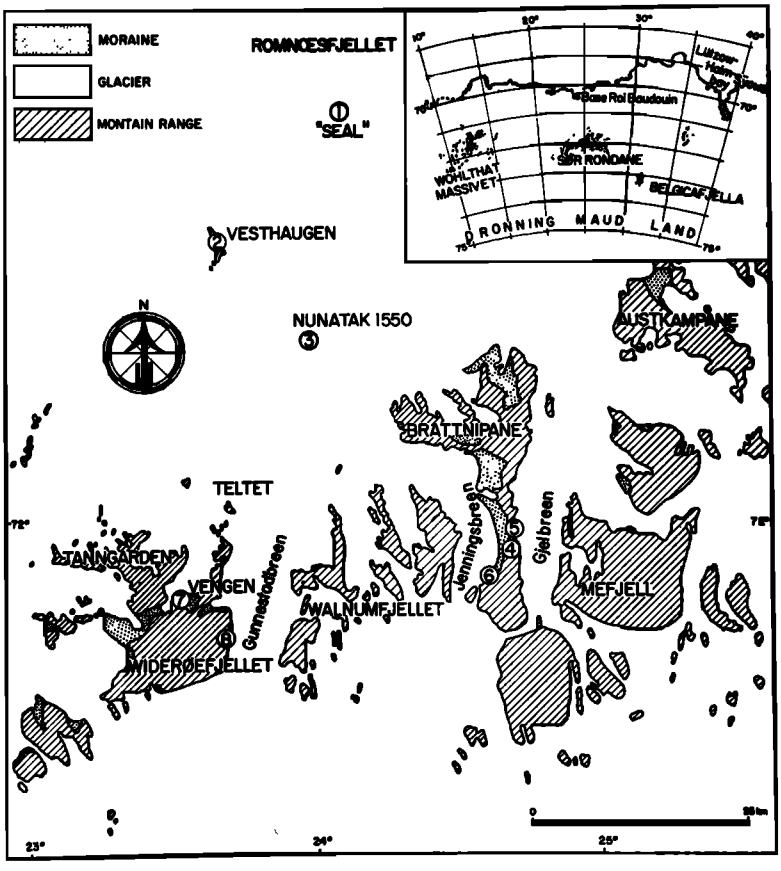


Fig. 1. Map of the western Sør Rondane Mountains showing the sampling sites. Lunckerygge is the mountain ridge indicated by sampling sites 4, 5, and 6.

during the period of collection. All azimuthal directions in this paper are given in degrees east of true north. Only the demagnetization diagrams are not corrected for the local present-day geomagnetic declination.

Results

The measurements and analyses of the NRM (natural remanent magnetization) of the samples led to a division into at least three groups:

1. Some rocks proved to be unsuitable for the paleomagnetic investigation: the granite from Lunckerygge, the granite dykes in the syenite and in the granite from Lunckerygge, the granite from Vengen, and the metamorphosed diabase from Widerøefjellet. These rocks have a relatively high magnetic susceptibility but only a relatively weak magnetic remanence. This relation can be given by the proportion of the NRM to the total of both magnetic remanence and induced magnetization, the latter in a magnetic field being equivalent to the geomagnetic field. For the rocks under discussion this 'proportional Q value' is not larger than 15% and is generally less than 10% with a minimum of 2%. The directions of the relatively weak NRM of these rocks, shortly after arrival and preparation, were close to the local present geomagnetic direction in the Sør Rondane region (see Figure 2). The scatter appeared to be clearly a consequence of their viscous nature, because repeated measurements after some months' storage in the laboratory revealed appreciable changes toward the new ambient field direction, in some cases even outrageous changes. The supposition that the NRM of these rocks is entirely a viscous remanence is redundantly supported by their progressive ac demagnetizations: this NRM was largely or

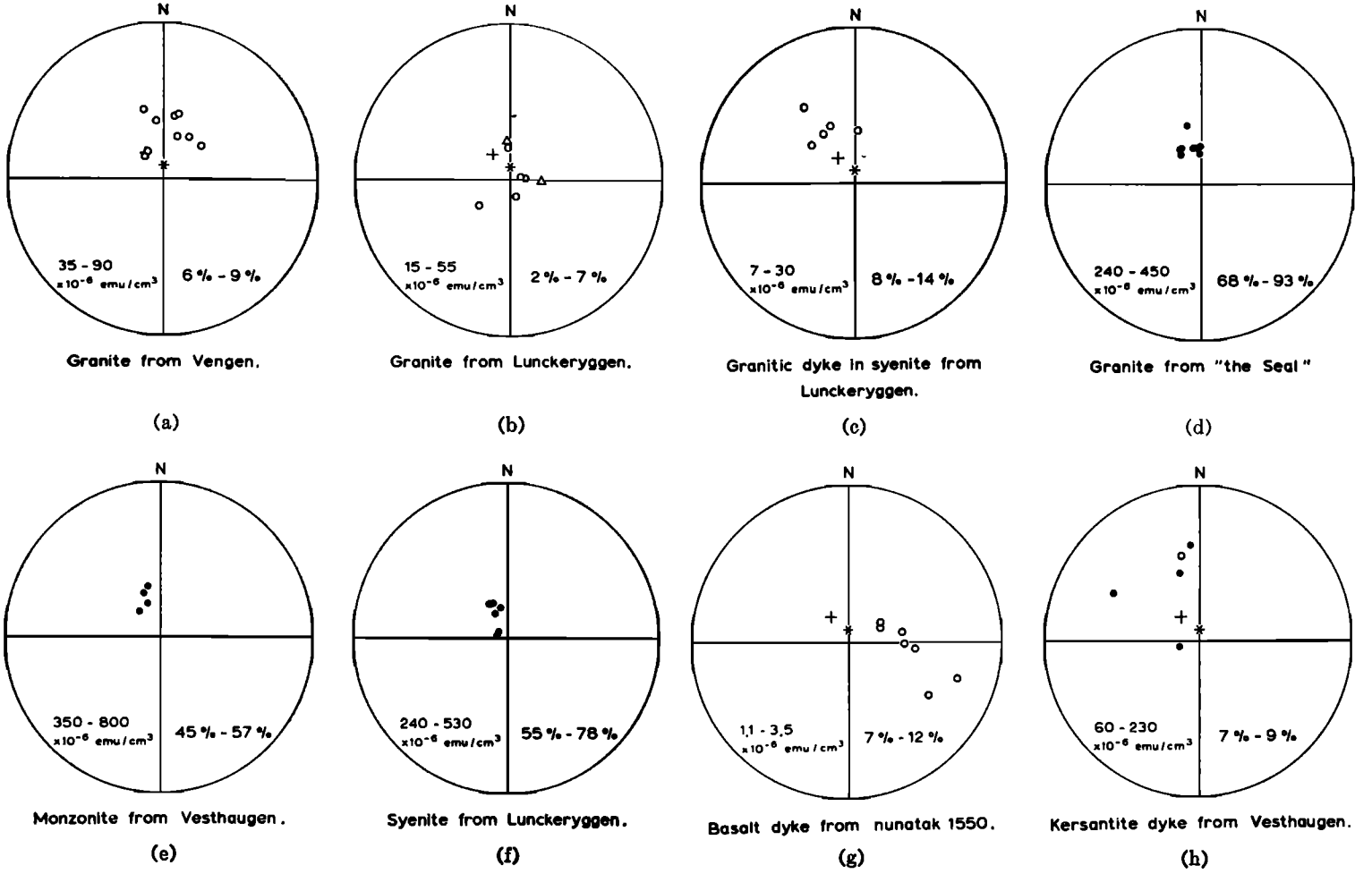


Fig. 2. Results of initial measurements of the NRM of the Sør Rondane igneous rocks. North-seeking directions of magnetization are plotted; stereographic projections; plane of projection is horizontal. Solid symbols denote north-seeking poles pointing downward, and open symbols denote north-seeking poles pointing upward. An asterisk indicates direction of axial dipole field (pointing upward), and a cross is direction of actual local geomagnetic field (pointing upward). Triangles in stereogram for granite from Lunckerygge are directions of aplitic dyke in this granite. On the left-hand side under in each stereogram are given the weakest and strongest intensity of the NRM observed in the samples of the rock in question. On the right-hand side under in each stereogram are given the minimum and maximum 'proportional Q value,' i.e. the proportion of NRM to the combined total of NRM and induced magnetization.

even entirely eliminated in ac magnetic fields as low as 100 to 200 oe.

2. More promising was the NRM of the granite from the Seal nunatak, the monzonite from Vesthaugen, and the syenite from Lunckerygge. The intensity of the NRM of these rocks was 10 times stronger than that of the other sampled rocks. Moreover, their 'proportional Q value' was very large (the granite up to 93%).

The directions of this strong NRM have a large within-site consistency, are similar in all three intrusives, and are strongly divergent from the present field in the Sør Rondane Mountains. Progressive demagnetizations with ac magnetic fields showed that this NRM did indeed consist almost entirely of only one fairly hard remanent component (see Figure 3*a*). The lower ac magnetic fields eliminated a small anomalous component, resulting in a slight adjustment of the magnetization directions.

All samples from the three intrusives were carefully demagnetized with ac magnetic fields and the direction of the harder, characteristic remanent magnetization was deduced. The result is shown in Figure 3*b* and summarized in Table 2. It is clear that the characteristic NRM directions of the three intrusive rocks agree closely.

3. The results of the samples of the relatively young basaltic dykes from Vesthaugen and from nunatak 1550 are less consistent. The 'proportional Q-value' of their NRM was very low. Moreover, the NRM of the dyke from nunatak 1550 had a surprisingly low intensity.

The directions of the NRM of these dykes diverge, however, from each other and from the present geomagnetic field in the Sør Rondane Mountains (the samples from nunatak 1550 display a 'streaking' toward the present field). Some progressive ac magnetic field demagnetizations showed that this NRM was hard for the greater part. These ac magnetic field demagnetizations failed, however, to reveal a distinct characteristic magnetization direction, largely owing to a scatter in the results. Consequently, no approximations of the NRM directions of these rocks are attempted. We can only report that the ac demagnetization revealed that the directions of the NRM of the dyke from Vesthaugen tend to approach the characteristic direction of the previous group, to which its host rock also belongs, and that the directions of the NRM of the sill from nunatak 1550 tend to collect in the ESE corner with a moderate, negative inclination.

Discussion

The common NRM direction of the granite from the Seal, the monzonite from Vesthaugen, and the syenite from Lunckerygge seems to be characteristic for the main igneous complex. This NRM cannot, of course, be older than the 530 m.y. inferred by *Picciotto et al.* [1964, 1966] from the zircon ages as the time of emplacement of the intrusives. The characteristic NRM most likely dates from the final thermal event, however, which is recorded by the Rb-Sr age of the biotites. Heating experiments with our samples showed that the characteristic NRM decayed over a range of temperatures up to 450°–575°C, which corresponds to the range of temperatures in which the loss of radiogenic strontium from biotite during regional heating occurs [*Hanson and Gast*, 1967]. The age of the characteristic NRM is therefore most likely given by the Rb-Sr age of the biotites and thus is approximately 480 m.y., which corresponds with the Lower Ordovician [*Kulp*, 1961]. From the paleomagnetic point of view it is unimportant whether the final thermal event was the regional mesozonal retrograde metamorphism, as suggested by *Michot and Klerkx* [1966], or the actual emplacement of the intrusives.

Although a post-intrusive tilting is unlikely (E. Picciotto and J. Michot, personal communication, 1966), the possibility could not be excluded with certainty. *van Autenboer* [1964] reported that the large-scale structure tends east–west and dips 40° to 50° to the south. From data in a more recent paper [*van Autenboer and Loy*, 1966] it can be inferred that in the mountains on both sides of the Gunnestadbreen (Teltet, Vengen, Widerøefjellet, Walnumfjellet) the orientation of the gneiss dips at 45° to 75° uniformly to the south (dip azimuth varies between N 175°E and N 195°E); this dip increases from north to south. In the region of Brattnipane this orientation dips 25° to 40° in a more westerly direction (dip azimuth varies between N 205°E and N 225°E).

Knowledge of the structural setting of the nunataks to the north of the mountain range is poor. The only details reported are that at

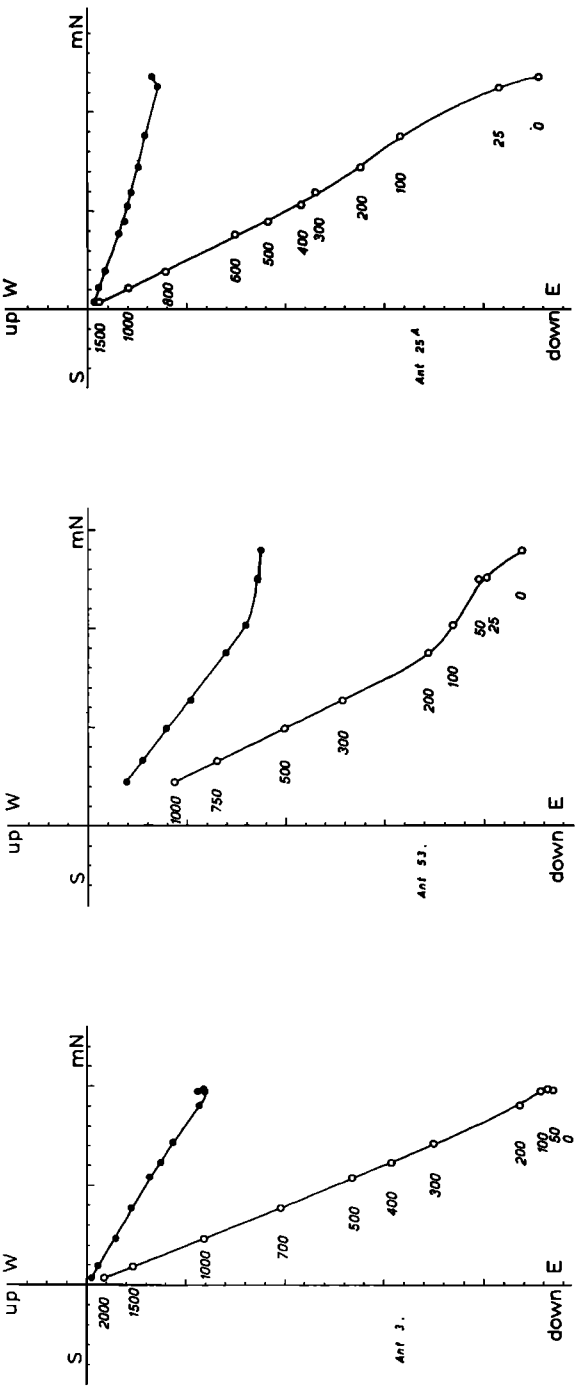


Figure 3a.

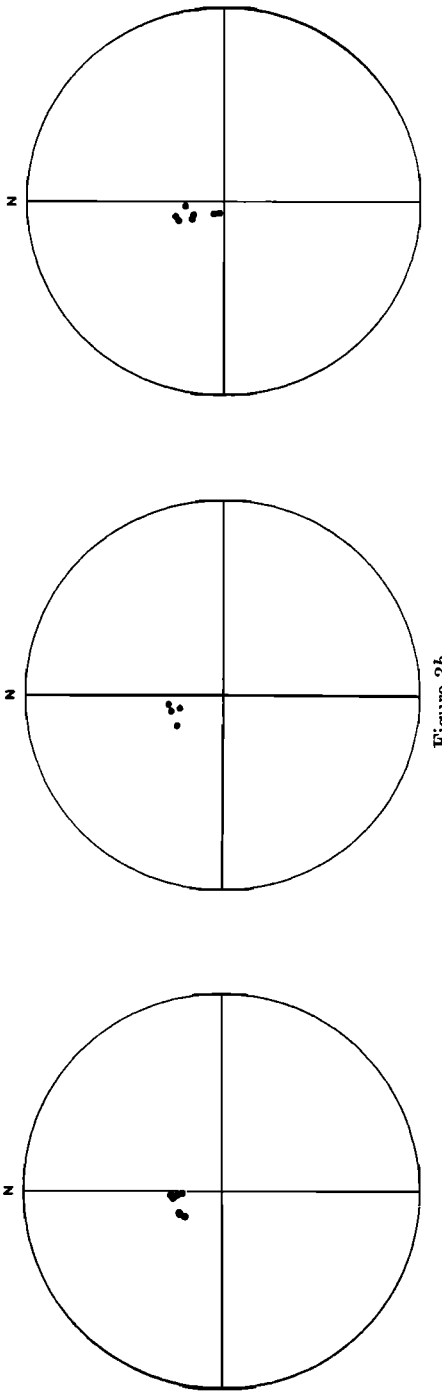


Figure 3b.

Fig. 3a. (*Opposite page.*) Demagnetization diagrams of one sample each from the granite from the Seal, the monzonite from Vesthaugen, and the syenite from Lunckerygge (left to right). The plotted points in such demagnetization diagrams represent successive positions, in orthogonal projection, of the end of the resultant magnetization vector during progressive demagnetization. Solid symbols represent projections on the horizontal plane; open symbols, projections on the north–south vertical plane. Numbers denote ac magnetic field intensities in oersted. Each unit on either axis represents 14×10^{-6} emu/cm^3. These diagrams are not corrected for the local present-day geomagnetic declination.

Fig. 3b. (*Opposite page.*) Stereograms with the directions of the hard NRM component of the granite from the Seal, the monzonite from Vesthaugen, and the syenite from Lunckerygge (left to right). For explanation see Figure 2.

Vesthaugen the orientation of the gneiss dips at approximately 30°S, and that the granite of Romnæsfjellet 'se débite en bancs épais' (probably sheet jointing) having a general east–west direction and a dip of 50°N [*Picciotto et al.*, 1966]. There is, however, no knowledge on whether or not all these structures have a relation to any tilting after the emplacement of the intrusives.

On the other hand, the close consistency of the paleomagnetic directions from the three widely spaced intrusives excludes the possibility of larger, mutually different, posterior tilting.

Moreover, the characteristic direction of these Sør Rondane intrusives appeared to be remarkably consistent with the paleomagnetic directions of the gneiss from the Ongul Island and the coast of Lützow-Holm Bay, some 650 km northeast of our region [*Nagata and Shimizu*, 1959; *Nagata and Yama-ai*, 1961]. These gneisses belong to the same metamorphic complex as the gneisses of the Sør Rondane Mountains. *Nicolaysen et al.* [1961] obtained Rb-Sr ages varying between 500 and 530 m.y. on large biotite crystals from three granitic pegmatites and a basic charnockite lens occurring in the Lützow-Holm Bay region. For unspecified reasons Nagata corrected his data for an attitude of the banded structure of the gneiss, having a dip of 50° and a strike of N 10°W.

A comparison of the characteristic NRM directions in situ of the gneisses of Lützow-Holm Bay and the Ongul Island with the NRM direction of the igneous rocks from the Sør Rondane Mountains (Tables 3 and 2, respectively) shows, however, that the directions are remarkably similar, whereas a correction for the 50° south dip of the structure in the western Sør Rondane Mountains and for the 50° east dip of the structure around Lützow-Holm Bay would result in an appreciable scatter of both results. Thus, we think that the paleomagnetic data from both regions indicate that (1) The NRM of the gneisses from the Lützow-Holm Bay area, just as the NRM of the igneous rocks of the Sør Rondane Mountains, dates from the last metamorphic event in the area, and (2) That no appreciable movements have taken place since this final metamorphic episode. The character-

TABLE 3. Mean Directions of the Hard NRM of the Reversed Magnetized Sample Gneisses from Ongul Island and Lützow-Holm Bay [*Nagata and Yama-ai*, 1961]

Locality	Number of Specimens	Direction of Magnetization	
		D	*I*
East Ongul Island (1957–1958)	18	−24°	58°
East and west Ongul Island (1959–1960)	5	−10°	49°
East coast of Lützow-Holm Bay	7	−19°	60°

TABLE 2. Mean Directions of the Hard NRM of the Reversed Magnetized Samples of Igneous Rocks from Sør Rondane

Mean characteristic magnetization direction of these igneous rocks showed: number of localities, 3; mean direction of characteristic NRM: $D = 341\frac{1}{2}°$ and $I = +64°$: $\alpha_{95} = 4\frac{1}{2}°$.

Locality	Rock Type	Number of Samples	Mean Direction of Characteristic NRM		
			D	*I*	α_{95}
The Seal	Granite	7	344°	+64½°	4½°
Vesthaugen	Monzonite	4	340°	+61°	6½°
Lunckerygge	Syenite	5*	340°	+66½°	1½°

* Two adjacent samples with somewhat deviating directions were omitted. For all seven samples the $\alpha_{95} = 7\frac{1}{2}°$ and the mean direction $D = 336°$ and $I = +71°$.

TABLE 4. Ancient Pole Positions Derived from Paleozoic Rocks of Gondwanic Continents

Italicized items are early data that have not been tested by demagnetization studies and thus must be regarded as provisional.

No.	Locality	Formation	Number of Samples and Sites*	Direction of Characteristic Magnetization D	I	α_{95}*	Pole Position	Author
		ANTARCTICA						
1	69°S, 39.5°E	Lützow-Holm Bay gneisses (Lower Ordovician)	30 (≥5)	340°	+58°	†	19°S, 23°E	Nagata and Yama-ai (1961)
2	72°S, 24°E	Sør Rondane igneous intrusions (Lower Ordovician)	16 (3)	341.5°	+64°	(4.5°)	28.5°S, 9.5°E	Zijderveld
		AUSTRALIA						
3	*16°S, 126°E*	*Antrim plateau basalts (Lower Cambrian)*	*7 (3)*	*53°*	*−2°*	*12°*	*36°S, 26°E*	*Irving and Green (1958)*
4	*16°S, 126°E*	*Elder Mountain sandstone (Middle Cambrian)*	*3*	*231°*	*−15°*	*10°*	*34°S, 16°E*	*Irving (1964)*
5	*34.5°S, 149°E*	*Duro porphyry (Middle Silurian)*	*7 (2)*	*306°*	*−11°*	*18°*	*32°S, 104°W‡*	*Irving (1964)*
6	35°S, 149.5°E	Mugga Mugga porphyry (Upper Silurian) 500°C,	17 (1)	2°	−43°	7°	80°S, 20°W§	Briden (1966)
(6)		300 oe	7 (1)	22°	−44°	12.5°	69°S, 38°E	Briden (1966)
7	*35°S, 149°E*	*Canberra igneous rocks (Upper Silurian–Lower Devonian)*	*40 (8)*	*12°*	*−37°*	*(21°)*	*71°S, 8°E*	*Irving (1964)*
8	*34.5°S, 149°E*	*Murrumbidgee red sediments (Middle Devonian)*	*11 (1)*	*40°*	*−29°*	*10°*	*49°S, 41°E*	*Irving (1964)*
9	*37°S, 150°E*	*Yalwal stage basalts (Upper Devonian)*	*19 (7)*	*5°*	*−23°*	*(14°)*	*65°S, 20°W*	*Irving (1964)*
10	32.5°S, 151.5°E	Lower Kuttung volcanic stage, with respect to bedding	8 (3)	154°	+45°		67°S, 110°W¶	Irving (1966)
(10)		(Lower Carboniferous) with respect to horizontal		187°	+50°		84°S, 46°E¶	Irving (1966)
11	32.5°S, 151.5°E	Upper Kuttung Paterson toscanite flow (Upper Carboniferous)	12 (4)	2°	−67°	(3°)	73°S, 147°E	Irving (1966)
12	30°–32.5°S, 150.5°–151.5°E	Upper Kuttung Main glacial stage (Upper Carboniferous)	83 (18)	201°	+80°	(5°)	50°S, 140°E	Irving (1966)
13	32.5°S, 151.5°E	Lower Marine basalt flow (Lower Permian)	6 (1)	230°	+76°		46°S, 122°E	Irving and Parry (1963)
14	34.5°S, 151°E	Upper Marine latite flows (Middle–Upper Permian)	43 (11)	232°	+81°	(6°)	44°S, 132°E	Irving and Parry (1963)
		SOUTH AMERICA‖						
15	23°S, 66°W	Cambrian redbeds	17	17°	+56°	14°	27°N, 50°W	Creer (1965)
16	19°S, 58°W	Urucum arkoses (Silurian)	23	37°	+41°	9°	34°N, 16°W	Creer (1965)
17	23°S, 66°W	Devonian redbeds	10	56°	+51°	8°	13°N, 19°W‡	Creer (1965)
18	17°S, 65°W	Taiguati formation (Pennsylvanian)	38	294°	−72.5°	5°	28°S, 34°W	Creer (1965)
19	5°S, 43°W	Piaui formation (Pennsylvanian)	20	165°	+44°	9°	65°S, 10°W‡	Creer (1965)
20	30°S, 68°W	Permian redbeds	53	160°	+66°	4°	66°S, 34°W‡	Creer (1965)
		AFRICA						
21	*34°S, 18°E*	*Cape system redbeds (Silurian)*	*8 (1)*	*162°*	*−4°*	*5°*	*50°N, 11°W*	*Irving (1964); Graham and Hales (1961)*
22	*18°S, 29°E*	*Dwyka varved clay (Upper Carboniferous)*	*4 (1)*	*310°*	*+86°*		*15°S, 23°E*	*Nairn (1960); Creer (1965)*
23	9.5°S, 34°E	Ecca redbeds (Lower Permian)	61 (9)			(11°)	38°S, 65°E	Opdyke (1964)

* Items included in parentheses refer to the sites.

† Mean of NRM directions from Ongul Island and Coast of Lützow-Holm Bay, respectively (see Table 2). In calculating the mean NRM direction (by direction cosines) for the Ongul Island, the reported determinations have been weighted as follows: east Ongul (1957–1958), 3; east and west Ongul (1959–1960), 1.

‡ Pole positions different from those in original paper [*Irving*, 1964; *Creer*, 1965] after recalculation.

§ In original paper [*Briden*, 1966] longitude of this pole erroneously given as 159.7°W instead of the correct 159.7°E.

¶ In original paper [*Irving*, 1966, p. 6041] these two pole positions have been interchanged.

‖ These South American data are from *Creer* [1965] and were taken without considering the small differences in location coordinates and ages found in various other papers by this author.

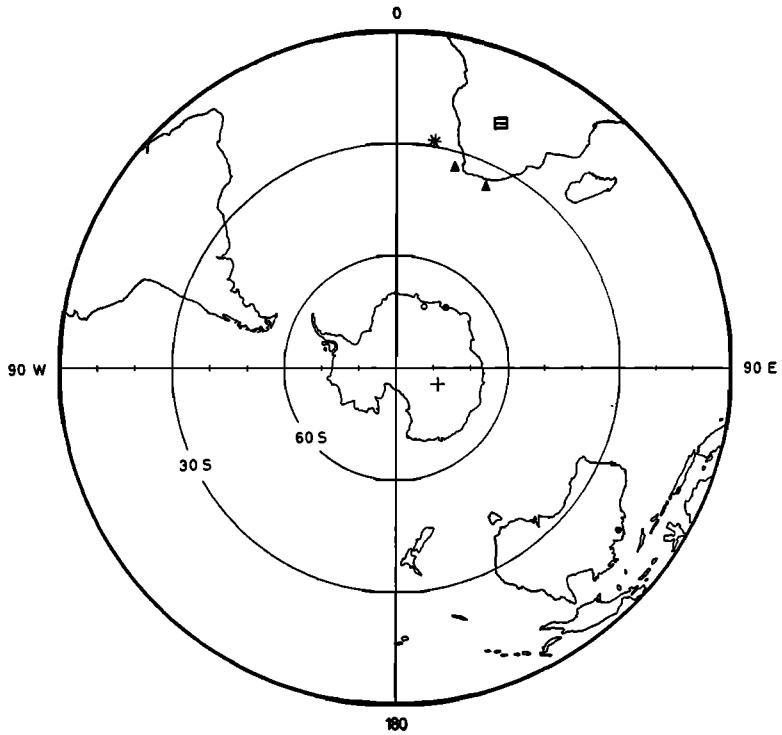


Fig. 4. Position of the paleomagnetic pole (asterisk) for the mean NRM direction of some intrusives of the Sør Rondane Mountains (circle). The mean paleomagnetic pole (double box) for the NRM directions of the gneisses from the Ongul Island and Lützow-Holm Bay (full dot on Antarctica) and both paleomagnetic poles (triangles) for the NRM directions of the Lower Cambrian Antrim basalts and the Middle Cambrian Elder Mountain sandstone of Australia (full dot on this continent) are also plotted. Cross marks the present geomagnetic pole.

istic NRM direction of the rocks in situ may thus represent the direction of the local geomagnetic field during the Upper Cambrian or Lower Ordovician.

Antarctica and Other Gondwanic Continents in Paleozoic Era

Since only a few Paleozoic paleomagnetic data are available for the Gondwanic continents (Table 4), no definite discussion of our Antarctic result is possible. Nevertheless, some general remarks might be of importance for paleogeographic reconstructions of Gondwanaland.

Creer [1964, 1965] showed that, when South America and Africa are placed adjacent to one another in their classical fit, the few Paleozoic poles for Africa match his Paleozoic polar wandering curve for South America. The paleomagnetic data thus confirmed this reconstruction, and, moreover, the outline of the Paleozoic polar wandering curve for Gondwanaland was thereby indicated. The tentative Paleozoic polar wandering curve for Australia [*Irving*, 1964] has, however, a rather different shape. It is not possible to match these two Gondwanic polar wandering curves satisfactorily (Figures 5 and 6). The length of the upper Paleozoic Australian polar wandering curve is appreciably shorter than the contemporaneous part of the South American-African curve. This difference might be explained by a drifting apart of Australia and South America-Africa in the late Paleozoic. The lower Paleozoic reconstruction is then obtained by bringing together the Silurian-Devonian poles of Australia (6, 7, 8, and 9) and South America-Africa (16, 21, and 17). In Figure 5 it is shown that, following this interpretation, Australia was situated close to the (present

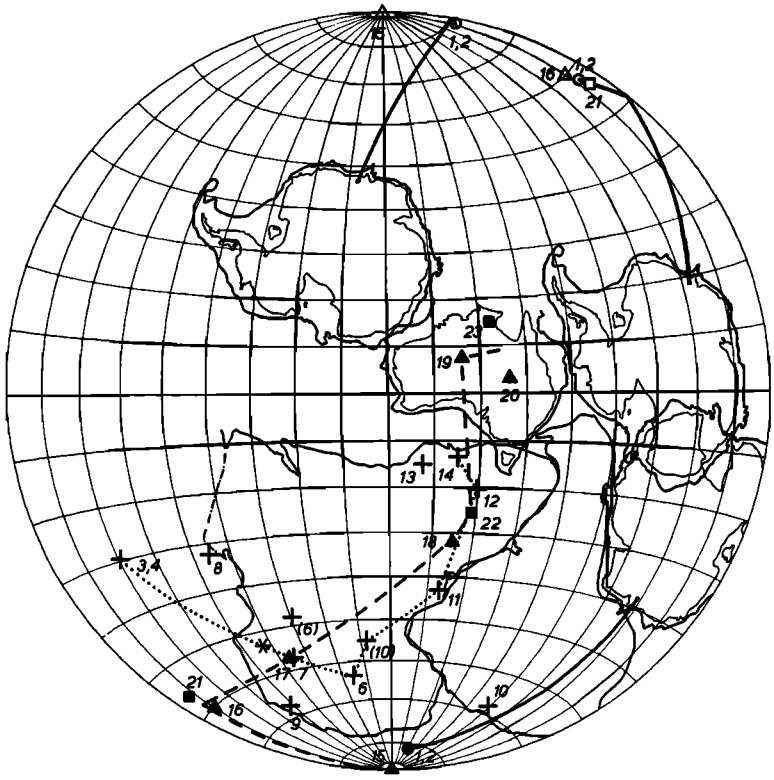


Fig. 5. Reassemblage of Gondwanaland showing only three of the various possible positions of Antarctica on account of the mean of present Lower Ordovician data. The paleomagnetic reconstruction of South America-Africa and Australia, obtained by fitting the Australian polar wandering curve [*Irving*, 1964; *Irving*, 1966] into the common South American-African polar wandering curve [*Creer*, 1965], shows clearly the dissimilarity of both curves and, moreover, their ambiguity in the lower Paleozoic. The continents are indicated with their —1000-fathom contours, and, for clarity, the present coast lines of Antarctica and Australia are also drawn. Index numbers of the paleomagnetic poles refer to Table 4. Open pole symbols are opposite the solid pole symbols. Asterisk is average paleomagnetic pole for the Devonian of Australia [*Irving*, 1964]. Heavy lines from Antarctica are colatitude parts of paleomeridians over the mean locality of the Sør Rondane mountains and Lützow-Holm Bay for the Lower Ordovician.

east) African coast during the lower Paleozoic, leaving no place for Antarctica there. In such a reconstruction it is not possible to settle the precise position of Antarctica from the present paleomagnetic data. The antarctic continent could have been situated somewhere along the Australian-South American outer border in Gondwanaland, against the (present north) coast of Australia as well as along the Chilean coast of South America (Figure 5). In the latter situation Antarctica appears as the lost 'Juan Fernandez Land' of the Chilean geologists [*Illies*, 1960].

The lower Paleozoic conditions are, however, the more ambiguous, since the South American and Australian paleomagnetic data seem to indicate polar wandering curves in opposite directions (Figure 5). This might suggest the alternative assumption that the South American and the Australian polar wandering curves represent polar wandering curves on opposite sides of the globe. In that case, no large Gondwanaland existed, and, for instance, the Australian upper Paleozoic glaciations were antipodal to the South African upper Paleozoic glaciations. It should be kept in mind, however, that, although the upper Paleozoic data of Australia (poles 10, 11, 12, 13, and 14) are based on

observations of NRM with cleaning and thus are reliable, there unfortunately seems to be some uncertainty as to the reliability of the lower Paleozoic data from Australia. *Irving* [1966] remarked that his Devonian results (poles 7, 8, and 9) are 'early data based on observation of NRM which require reinvestigation.' Recent investigations of the ancient NRM direction in the Mugga Mugga porphyry, one of the Canberra intrusives, yielded two results, depending on whether the samples were cleaned magnetically or thermally [*Briden*, 1966]. Moreover, nothing is reported about subsequent tectonical influences.

Although both Australian Cambrian poles (3 and 4) belong to the 'early data,' they coincide very well with our antarctic Lower Ordovician poles. Already in the present arrangement of the continents these early Paleozoic poles come close together near South Africa (Figure 4). When Antarctica is fitted in, as is usually done [*Du Toit*, 1937; *Carey*, 1956; *King*, 1962; *Wilson*, 1963], with its convex Wilkesland coast in the concave south coast of Australia, the mean of both antarctic poles coincides with the mean of both Australian poles. This could be taken as a support for this common Antarctica-Australia reconstruction.

Figure 6 is an attempt to synthesize all the present data in one single Gondwanaland with one Gondwanic polar wandering curve. This hypothetical but satisfactory reconstruction of Figure 6 is obtained by retaining both classical

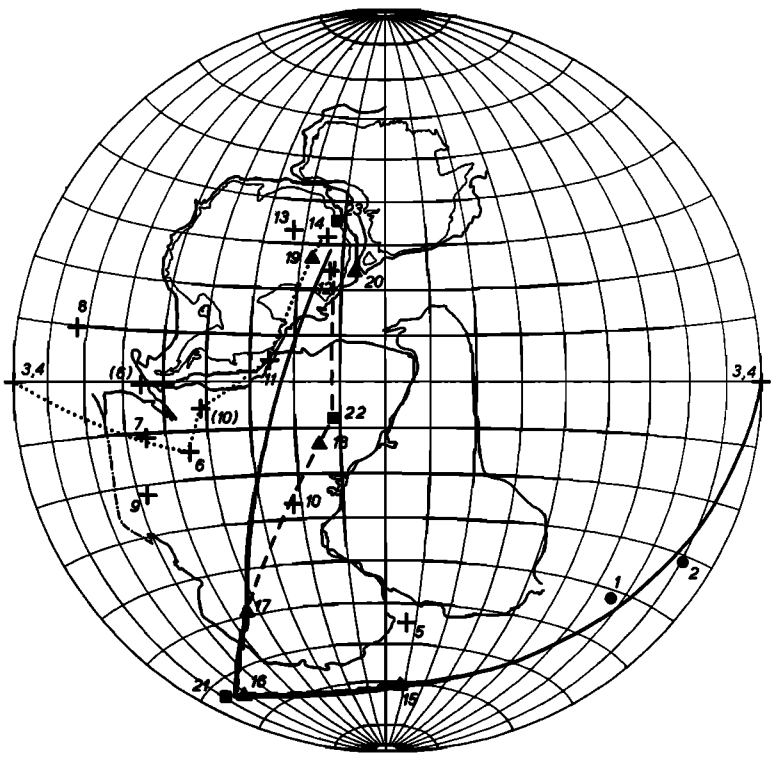


Fig. 6. A proposed reconstruction of the four larger Gondwanic continents for the Paleozoic, obtained by placing Australia with its upper Paleozoic poles near the coeval South American and African poles and Antarctica with its Wilkesland coast adjacent to the south coast of Australia. The continents are indicated with their —1000-fathom contours, and, for clarity, the present coast lines of Antarctica and Australia are also shown. Index numbers of the paleomagnetic poles refer to Table 4. The dotted curve is the usually accepted Australian polar wandering curve [*Irving*, 1964; *Irving*, 1966]; the dashed curve is the common South American-African polar wandering curve [*Creer*, 1965]; the continuous curve is the proposed Gondwanic polar wandering curve. In this reconstruction the Australian paleomagnetic data of Devonian rocks from the Canberra region remain inexplicable.

reconstructions of South America with Africa and of Antarctica with Australia, respectively and, moreover, by matching the reliable upper Paleozoic Australian poles to the South American-African upper Paleozoic polar wandering curve. An indication of the lower Paleozoic polar wandering curve shows up. The great length of this Cambro-Ordovician polar wandering curve is not unacceptable, if we realize that both periods represent half of the Paleozoic era [*Kulp*, 1961]. In fact, in such a reconstruction it would appear that the mean rate of apparent polar wandering was about 75 km in 1 m.y. for both the lower and the upper Paleozoic.

The pole deduced from the Cambrian redbeds of South America is situated on the Upper Ordovician part of the polar wandering curve. Here one should realize that, when such a common polar wandering curve existed for the Gondwanic continents, all their Paleozoic poles should after reconstruction closely approximate this curve irrespective of their correct age and that variance between the age of the pole and its place on the curve could be simply a consequence of a misinterpretation of the age of the NRM in the rocks in question. This might also be an explanation for the dissentient paleomagnetic data from the Devonian rocks in the Canberra region. In fact, these Canberra Devonian poles are situated outside the sketched Gondwanic polar wandering curve. It should be realized, however, that the paleomagnetic data of the Gondwanic continents are very indigent and one can thus expect refinements and modifications of the proposed reconstruction and polar wandering curve in the future.

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